

Cd $\phi_2$   $\frac{11-12}{2}$  →

$$\begin{aligned}
& \frac{1}{16\pi^2} \left( -\frac{1}{18} g_3 g_1^2 y_d^{i1i2} LF_{2,1,0}[m_1, m_d^{i2}] + \frac{1}{9} g_3 g_1^2 y_d^{i1i2} LF_{3,1,-1}[m_1, m_d^{i2}] - \frac{1}{18} g_3 g_1^2 y_d^{i1i2} \right. \\
& LF_{4,1,-2}[m_1, m_d^{i2}] - \frac{1}{72} g_3 g_1^2 y_d^{i1i2} LF_{2,1,0}[m_1, m_q^{i1}] + \frac{1}{36} g_3 g_1^2 y_d^{i1i2} LF_{3,1,-1}[m_1, m_q^{i1}] - \\
& \frac{1}{72} g_3 g_1^2 y_d^{i1i2} LF_{4,1,-2}[m_1, m_q^{i1}] - \frac{3}{8} g_3 g_2^2 y_d^{i1i2} LF_{2,1,0}[m_2, m_q^{i1}] + \\
& \frac{3}{4} g_3 g_2^2 y_d^{i1i2} LF_{3,1,-1}[m_2, m_q^{i1}] - \frac{3}{8} g_3 g_2^2 y_d^{i1i2} LF_{4,1,-2}[m_2, m_q^{i1}] + \\
& \frac{1}{12} g_3^3 y_d^{i1i2} LF_{2,1,0}[m_3, m_d^{i2}] + \frac{7}{12} g_3^3 y_d^{i1i2} LF_{3,1,-1}[m_3, m_d^{i2}] - \\
& \frac{2}{3} g_3^3 y_d^{i1i2} LF_{4,1,-2}[m_3, m_d^{i2}] + \frac{1}{12} g_3^3 y_d^{i1i2} LF_{2,1,0}[m_3, m_q^{i1}] + \\
& \frac{7}{12} g_3^3 y_d^{i1i2} LF_{3,1,-1}[m_3, m_q^{i1}] - \frac{2}{3} g_3^3 y_d^{i1i2} LF_{4,1,-2}[m_3, m_q^{i1}] - \\
& \frac{1}{4} g_3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{2,1,0}[\tilde{\mu}, m_d^r] + \frac{1}{2} g_3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{3,1,-1}[\tilde{\mu}, m_d^r] - \\
& \frac{1}{4} g_3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{4,1,-2}[\tilde{\mu}, m_d^r] - \frac{1}{2} g_3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{2,1,0}[\tilde{\mu}, m_q^p] + \\
& g_3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{3,1,-1}[\tilde{\mu}, m_q^p] - \frac{1}{2} g_3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{4,1,-2}[\tilde{\mu}, m_q^p] - \\
& \frac{1}{4} g_3 y_d^{pi2} \overline{y_u^{pr}} y_u^{i1r} LF_{2,1,0}[\tilde{\mu}, m_u^r] + \frac{1}{2} g_3 y_d^{pi2} \overline{y_u^{pr}} y_u^{i1r} LF_{3,1,-1}[\tilde{\mu}, m_u^r] - \\
& \frac{1}{4} g_3 y_d^{pi2} \overline{y_u^{pr}} y_u^{i1r} LF_{4,1,-2}[\tilde{\mu}, m_u^r] - \frac{1}{18} g_3 m_1 g_1^2 a_d^{i1i2} LF_{2,1,1,0}[m_1, m_d^{i2}, m_q^{i1}] + \\
& \frac{1}{18} g_3 m_1 g_1^2 a_d^{i1i2} LF_{3,1,1,-1}[m_1, m_d^{i2}, m_q^{i1}] - \frac{1}{6} g_3 g_1^2 y_d^{i1i2} LF_{1,1,1,0}[m_1, m_d^{i2}, \tilde{\mu}] + \\
& \frac{1}{3} g_3 g_1^2 y_d^{i1i2} LF_{2,1,1,-1}[m_1, m_d^{i2}, \tilde{\mu}] - \frac{1}{6} g_3 g_1^2 y_d^{i1i2} LF_{3,1,1,-2}[m_1, m_d^{i2}, \tilde{\mu}] - \\
& \frac{1}{12} g_3 g_1^2 y_d^{i1i2} LF_{1,1,1,0}[m_1, m_q^{i1}, \tilde{\mu}] + \frac{1}{6} g_3 g_1^2 y_d^{i1i2} LF_{2,1,1,-1}[m_1, m_q^{i1}, \tilde{\mu}] - \\
& \frac{1}{12} g_3 g_1^2 y_d^{i1i2} LF_{3,1,1,-2}[m_1, m_q^{i1}, \tilde{\mu}] - \frac{1}{6} g_3 g_1^2 y_d^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_d^{i2}] - \\
& \frac{1}{12} g_3 g_1^2 y_d^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_q^{i1}] - \frac{3}{4} g_3 g_2^2 y_d^{i1i2} LF_{1,1,1,0}[m_2, m_q^{i1}, \tilde{\mu}] + \\
& \frac{3}{2} g_3 g_2^2 y_d^{i1i2} LF_{2,1,1,-1}[m_2, m_q^{i1}, \tilde{\mu}] - \frac{3}{4} g_3 g_2^2 y_d^{i1i2} LF_{3,1,1,-2}[m_2, m_q^{i1}, \tilde{\mu}] - \\
& \frac{3}{4} g_3 g_2^2 y_d^{i1i2} LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_q^{i1}] - \frac{1}{6} m_3 g_3^3 a_d^{i1i2} LF_{2,1,1,0}[m_3, m_d^{i2}, m_q^{i1}] - \\
& \frac{4}{3} m_3 g_3^3 a_d^{i1i2} LF_{3,1,1,-1}[m_3, m_d^{i2}, m_q^{i1}] + \frac{1}{3} g_3 g_1^2 y_d^{i1i2} LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_d^{i2}] - \\
& \frac{1}{6} g_3 g_1^2 y_d^{i1i2} LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_d^{i2}] + \frac{1}{6} g_3 g_1^2 y_d^{i1i2} LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_q^{i1}] - \\
& \frac{1}{12} g_3 g_1^2 y_d^{i1i2} LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_q^{i1}] + \frac{3}{2} g_3 g_2^2 y_d^{i1i2} LF_{2,1,1,-1}[\tilde{\mu}, m_2, m_q^{i1}] - \\
& \frac{3}{4} g_3 g_2^2 y_d^{i1i2} LF_{3,1,1,-2}[\tilde{\mu}, m_2, m_q^{i1}] + \frac{1}{2} g_3 \tilde{\mu}^2 y_d^{pi2} \overline{y_u^{pr}} y_u^{i1r} LF_{2,1,1,0}[\tilde{\mu}, m_q^p, m_u^r] - \\
& \left. \frac{1}{2} g_3 \tilde{\mu}^2 y_d^{pi2} \overline{y_u^{pr}} y_u^{i1r} LF_{3,1,1,-1}[\tilde{\mu}, m_q^p, m_u^r] \right)
\end{aligned}$$